RESEARCH CONTRIBUTION 23 OCTOBER 2026

Smaller Diophantine consistency equations

Pell saturation, exact transport elimination, and coprime proof-flow certificates

Prepared for Vladimir Reshetnikov and the ProveIt project.
AI-assisted mathematical research with explicit polynomial artifacts.

Abstract

We improve two different parts of the current ProveIt frontier. A transport coordinate change, already available in an earlier source but not applied to the newest joint-sign family, gives one polynomial in **18 unrestricted integers**, of exact degree **88**, evaluated in **105** binary arithmetic operations. Its predecessor has 19 variables and 109 operations. Three companion projections improve the variable counts at degrees 32, 20, and 16 as well.

A separate Pell-certificate reduction removes a redundant factor from its strong ordinate. The square root needed to restore that factor is already certified by another part of the same equation. This gives a **nine-positive-variable** polynomial of degree **1,628,768** in **434** operations, improving the current 440-operation, degree-1,845,940 construction in both measures. A complete-radical majorant gives another nine-variable point of degree **1,103,950** in **1,074** operations. All fixed coefficients and ordinary division-free evaluation circuits are supplied.

The new existence equivalences are proved with explicit reverse maps. Their consistency interpretation composes with the precisely pinned ordinary native/compiler and syntactic-refutation calibration theorems. We do not claim a new end-to-end Lean/Rocq certificate, a completed derivation in the repository's encoded object-language ZFC, or global minimality. A distinct Fermat-label proof-flow construction removes acyclicity and common-multiple parameters at the finite-certificate level; its final fixed-dimensional Diophantine closure remains open and is not counted as a finished equation.

Reviewed repository snapshot: `c1a102936aca437e9999530a0111e6ddab56dc6d`.
Both supplied repository directories were inspected, including all five archives then in `docs/incoming`. All comparisons are to this frozen snapshot, not to a moving branch. The local date is 23 October; the snapshot's commit timestamp is 24 October in UTC.

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1 The mathematical target and the actual improvement

1.1 A single fixed polynomial and a specified domain

Fix the recursive proof predicate of the syntactic ZFC calculus used in ProveIt. With $\bot$ its chosen contradictory sentence, write

$$\mathrm{Con}(\mathsf{ZFC}) \;:=\; \forall p \in \mathbb{N}\ \neg\,\mathrm{Proof}_{\mathsf{ZFC}}(p, \ulcorner \bot \urcorner).$$

The requested endpoint is an integer-coefficient polynomial P, with no remaining program or theory parameter, and a domain D for which

$$\mathsf{ZFC} \vdash \left(\neg \exists \mathbf{x} \in D^m\ P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

The two sides of this biconditional are not asserted separately. A positive solution, a natural solution, and an unrestricted integer solution are different assertions. In this report $\mathbb{N}$ includes zero.

All displayed intermediate symbols are polynomial abbreviations unless expressly declared as unknowns. Fixed powers have fixed nonnegative integer exponents. Fixed coefficients are given both as lossless integer bytes and by deterministic finite recipes. No coefficient depends on a consistency decision, a future search, or a choice of a refutation. The literal JSON circuits define ordinary polynomials even where their full monomial expansions would be inconvenient to print.

1.2 Measured results

Construction	Domain	Variables	Exact degree	Operations
Transport projection	$\mathbb{Z}$	18	88	105
Companion	$\mathbb{Z}$	19	32	108
Companion	$\mathbb{Z}$	20	20	111
Companion	$\mathbb{Z}$	21	16	114
Pell saturation, short bound	$\mathbb{Z}_{>0}$	9	1,628,768	434
Pell saturation, radical bound	$\mathbb{Z}_{>0}$	9	1,103,950	1,074

For each of the first four rows, its predecessor has one more integer variable, the same degree, and four more operations. The reduction is an exact integer-polynomial projection, so it applies to every source zero; it does not rely on a new restriction to specially chosen canonical zeros.

The short nine-positive construction reduces degree from 1,845,940 to 1,628,768, an improvement of approximately 11.76%, while also reducing 440 operations to 434. Its natural-number translation has the same degree and 441 operations. The complete-radical construction is a separate degree/circuit tradeoff; the inspected incoming radical candidate has degree 1,158,228 and 1,753 operations. The new point has lower degree and the measured circuit cost given above. The smaller ten-integer and nine-natural variable counts already present in the repository remain separate results: the 18-integer equation is useful for its much lower degree and short expression, not as a fewest-variable record.

An operation is a live binary addition, subtraction, or multiplication. Squaring and multiplication by a fixed scalar are charged. Reusing an already computed value and supplying a fixed integer leaf are free under the repository's convention. This is arithmetic circuit size, not bit complexity; the fixed coefficients are large. A sum of six residual squares is one polynomial equation, rather than six separate asserted equations.

1.3 What is new, and what is reused

The transport coordinate change itself is already explicit in source 73, *Reanchored sign constraints and adaptive Pell carriers*. The new result is its application to the newer joint-sign projection family, with complete equations, inverse maps, exact degrees, and paid circuits. The argument is elementary; the improved endpoint is the contribution.

The saturated positive Pell certificate changes the certificate itself. It exploits a square root already forced elsewhere, retains the necessary coprimality conditions, and replaces the old majorant. The incoming round-eleven Lucas construction uses a related saturation idea in a different unrestricted-integer certificate. We credit that precursor. The positive certificate, its interface proof, and its full-equation costs are established here. The subsequent complete-radical optimization also builds on the incoming radical-majorant methodology; it is supported by exact integer coefficient certificates, not an asymptotic expansion used as an inequality.

The Fermat-label appendix is a genuinely different source architecture relative to the inspected budget and height-register constructions. Its ingredients are classical. We claim its proved finite-certificate theorem and identify the remaining compiler costs; we do not claim historical priority or an unmaterialized final polynomial record.

1.4 Proof layers and formalization status

There are four distinct kinds of evidence in this package:

1. Complete ordinary proofs of the new projections, Pell saturation, radical bounds, and finite proof-flow equivalence.
2. Exact finite checks binding the literal equations to those proofs: polynomial identities, coefficient reconstruction, nonzero leading coefficients, topology, liveness, and arithmetic operation counts.
3. The inherited ordinary compiler/refutation calibration, pinned to immutable source data and retained proof notes. This supplies the link from the unchanged arithmetic source to ZFC refutations.
4. A possible future kernel-checked derivation of the entire composed sentence inside the repository's encoded ZFC calculus.

The new results establish the stated solvability reductions. Their Con(ZFC) interpretation follows by composition with the retained ordinary calibration theorems. This contribution does not independently reprove the entire inherited semantic chain or supply its unfinished kernel-checked internalization. The repository itself records unfinished native phase/global calibration and internal-model transport obligations; this package does not silently mark them complete. Its already checked generic consistency equation remains a separate formal checkpoint.

The elementary constructions use finite arithmetic, induction, Pell recurrences, and finite coded derivations. They are suitable for formalization in ZFC. To interpret (1) as an ordinary theorem, compose them with the named ordinary calibration proofs. To interpret it as a claim about a newly exported proof object in the repository's formal calculus would be stronger and is not justified by this delivery. Finite tests alone do not prove an unbounded consistency equivalence.

2 Removing one integer witness from the joint-sign family

The smallest-degree branch and the smallest-variable branch need different comparisons. Here we improve four recent equations over *unrestricted integers*. The principal endpoint is a single polynomial in eighteen integer variables, of exact degree 88, with 105 live binary arithmetic operations. Its direct

predecessor in the incoming joint-guards packet uses nineteen variables, degree 88, and 109 operations.

Incoming predecessor			This composition		
Variables	Degree	Operations	Variables	Degree	Operations
19	88	109	18	88	105
20	32	112	19	32	108
21	20	115	20	20	111
22	16	118	21	16	114

Every entry in this table has domain $\mathbb{Z}$; none is obtained by silently relabelling positive witnesses. All four new equations retain precisely the same six fixed numerals as their direct predecessors. These are new compositions and measured endpoints, not a new general elimination lemma: source 73, *Guard reanchoring*, already uses the complement coordinate and monic transport elimination proved below. Its application there is to an earlier forty-one-variable system. The incoming joint-sign projection family retains the eliminable transport coordinate, which lets the same method produce the four improvements in the table.

2.1 An exact integer-polynomial projection

Suppose a source system contains polynomial abbreviations q, W independent of two integer coordinates F, α, and put $C = q - F - \alpha$. Suppose one of its rows is

$$(K_0 + W)C - F - z(q-1) = 0. \tag{2}$$

First replace α by the free integer coordinate C. This is an invertible polynomial change of coordinates, with inverse $\alpha = q - F - C$. The row is now monic in F. Define

$$\widehat{F} = (K_0 + W)C - z(q-1), \qquad \widehat{\alpha} = q - \widehat{F} - C. \tag{3}$$

Substitute these polynomials in every other row and delete Equation (2).

Proposition 2.1. *The old and new integer zero sets are in bijection. The projection fixes every other coordinate and sends (F, α) to $C = q - F - \alpha$; its inverse is Equation (3).*

Proof. At an old zero the transport row gives $F = \widehat{F}$, and then $\alpha = \widehat{\alpha}$. All substituted rows vanish. Conversely, restore these two values at a new zero. The deleted row vanishes identically, while every retained row vanishes by the defining substitution. Both compositions fix the original coordinates, because the old transport equation determines F uniquely. There is no division, positivity assumption, coprimality assumption, or new choice of canonical Pell witnesses in this argument. □

The rank expression also becomes shorter:

$$\begin{aligned}&((q-1)(q-F)+\alpha)(q^2-1) + (M_C + qM_F)J \\ &\qquad = (q(q-\widehat{F}) - C)(q^2-1) + (M_C + qM_F)J.\end{aligned} \tag{4}$$

This is a literal polynomial identity after the substitution, not merely an equality at solutions. The determinant guard remains valid with $F = \widehat{F}$ because it is substituted everywhere, including its sign constraints. No coefficient port is replaced or borrowed from another construction.

2.2 The complete eighteen-variable polynomial

The eighteen unknowns are exactly

$$J_3, C, z, f, h, r, v, w, \tau, y, \gamma, o, j, c, t, p_1, p_2, p_3 \in \mathbb{Z}. \tag{5}$$

The variable C is unrestricted; its required sign is enforced by the polynomial's inherited guard. All following symbols are polynomial abbreviations and do not add quantified variables. Define

$$\begin{aligned}
J &= 3J_3 + 1, & q &= \beta J + 1,\\
W &= 4(3w + 1), & X &= qW,\\
Y &= (2v + 1)q^3, & F &= (K_0 + W)C - z(q - 1),\\
E &= XY, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H,\\
V &= fo - c, & I &= (2r + 1)c^2,\\
k &= 1 - V + jc + hE, & \eta &= c - kY,\\
\zeta &= k - \eta, & T &= \eta\zeta.
\end{aligned}$$

Then put

$$\begin{aligned}
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
d &= (AF)(BC) - (q - AF - BC)^2,\\
P_+ &= 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{aligned}$$

Here $P_+ \geq 1$ for every integer triple; its standard surjectivity onto positive integers is used only in the inherited joint-sign theorem. The six residuals are

$$\begin{aligned}
L_1 &= \tau^2 - EkY(k + EkY) - 1,\\
L_2 &= (X + ac + \gamma H)^2 - \Delta c^2 - 1,\\
L_3 &= cj - V - R,\\
L_4 &= I^2 - \Delta(f^2 - 1),\\
L_5 &= I^2(V^2 - y^2) + y^2 - 1,\\
L_6 &= dT - (t^2 - d - T + 1)^2 - P_+.
\end{aligned} \tag{6}$$

The complete equation is

$$\boxed{Q_{18} = L_1^2 + L_2^2 + L_3^2 + L_4^2 + L_5^2 + L_6^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^{18}.} \tag{7}$$

The JSON uses the names `v_scale`, `w3`, `pair_root`, and `pair_pos1-3` for v, w, t, p_1, p_2, p_3. Those are only spelling changes. Its 105-gate straight-line program is an explicit division-free definition of the same ordinary polynomial.

2.3 The six fixed numerals

The tuple is the complete source-34 native binding named `mixed-parity-unrounded-1064921`, together with the two determinant scales. In the displayed equation,

$$\beta = 2^{1064921} - 1, \qquad A = 2^{1113}, \qquad B = 2^{666104}.$$

The other three fixed integers K_0, M_C, M_F are supplied as literal, compressed, minimal unsigned little-endian byte strings, with their complete finite construction retained alongside them. In particular M_F is the *already shifted* source port. It is not shifted again. All six numeral payloads together occupy 615883 bytes before compression.

For clarity, the finite construction of the native ports is as follows. The retained literal 215-state table determines 943 ordered Wang tiles by the explicit finite map in `integer/upstream/guards/upstream/source34/source34_builder.py`. With radix $\rho = 2^{121}$ and block length 8801, that map builds a 989-element support $\mathcal{E}$, its explicit nonnegative coefficients c_e, and the dictionary

$$d_s = \mathbf{1}_{s\in\{40,41,1100,2200\}} + \sum_{e\in\mathcal{E}} c_e\big(\mathbf{1}_{s=4393-e} + \mathbf{1}_{s=5495-e}\big).$$

Here the indicator sum counts coincident contributions. Put $D_C = \sum_s d_s\rho^s$ and $\mathcal{B} = \rho^{8801}$. The finite recipes then give

$$\begin{aligned}
K_0 &= D_C + \mathcal{B}\rho,\\
M_C &= \mathcal{B} - 1 - \sum_{e\in\mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063},\\
M_F &= (\mathcal{B}-1) + 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9}(\rho^{1000+2j} + \rho^{1021+2j})\\
&\qquad + \rho^{2195} + \rho^{3296} + (2^{119}-1)\sum_{e=8792}^{8798}\rho^e + (2^{27}-1)\rho^{8799},\\
\mu &= 1022 + \sum_{j=0}^{43} 2^{10+2j}.
\end{aligned}$$

The finite support, every tile and coefficient, and the generating table are included as data and executable finite recipes. This is not an appeal to an unspecified universal polynomial or a free compiler index. The verifier reconstructs the entire tuple, compares its integer values to the literal bytes, and replays 69230 finite convolution-basis checks. These finite checks bind the numerals; the unbounded semantic theorem is a separate inherited argument.

2.4 Why the joint row really enforces two positive signs

The inherited guard has a useful elementary proof, which also explains why the eighteen displayed inputs may range over all integers. For integers x, y, t put

$$G(x, y, t) = xy - (t^2 - x - y + 1)^2.$$

Then $x, y > 0$ if and only if $G(x, y, t) > 0$ for some integer t. Indeed, $G > 0$ implies $xy > 0$. If $x = -a, y = -b$ with $a, b > 0$, then $(t^2 + a + b + 1)^2 > ab$, a contradiction. Conversely, for positive x, y put $n = x + y - 1$. We have $xy \geq n$. If n is a square, take $t^2 = n$. Otherwise write $m^2 < n < (m + 1)^2$ and choose whichever of $m, m + 1$ makes $|t^2 - n|$ smaller. This distance is at most m, so $(t^2 - n)^2 \leq m^2 < n \leq xy$.

Every positive integer is a value of $1 + p_1^2 + p_2^2 + p_3(p_3 + 1)$. This follows from the three-square theorem applied to $4n - 3$: exactly one square is odd modulo four, giving the displayed expression after division by four. Consequently

$$x, y > 0 \quad\Longleftrightarrow\quad \exists t, p_1, p_2, p_3 \in \mathbb{Z} \quad G(x, y, t) = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

Apply this to $x = d, y = T$ in L_6. The new projection substitutes F in both occurrences of the determinant and in the rank; it does not delete this guard or infer positivity from the spelling of a variable. The further interpretation of $d > 0$ and $T > 0$ is exactly the retained parent's determinant and signed-Pell argument.

2.5 Correctness and its precise proof boundary

Let Q_{19}^{old} be the exact incoming polynomial `joint/joint19_degree88.json` in the archive `ZFC_Diophantine_Joint_Guards_2026-10-24.zip`. The repository is pinned at commit `c1a102936aca`; the complete identifier appears on the title page. Its source graph and the exact parent-to-polynomial binding are retained. The preceding projection gives the unconditional identity

$$Q_{18}(\mathbf{x}) = Q_{19}^{\mathrm{old}}(\mathbf{x}, F = \widehat{F}, \alpha = \widehat{\alpha}),$$

where the input correspondence replaces the old F, α by the new C. The deleted squared residual is identically zero. Consequently

$$\exists \mathbf{x} \in \mathbb{Z}^{18}\ Q_{18} = 0 \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Z}^{19}\ Q_{19}^{\mathrm{old}} = 0. \tag{8}$$

This equivalence is proved over arbitrary integer values of the six coefficient ports, so there is no hidden numerical or sign assumption in the new step.

For the authentic tuple the retained parent proof proceeds through the joint-positivity lemma, determinant sign restoration, the signed-power bootstrap, native Wang-tableau compilation, Pell calibration, and the literal machine/refutation theorem. Composing that ordinary proof chain with Equation (8) yields the intended theorem

$$\mathsf{ZFC} \ \vdash\ \left(\neg\exists \mathbf{x} \in \mathbb{Z}^{18}\ Q_{18}(\mathbf{x}) = 0\right) \ \longleftrightarrow\ \mathrm{Con}(\mathsf{ZFC}).$$

This statement neither assumes nor proves consistency. The new projection is an elementary argument formalizable in ZFC. The archive does not provide a new end-to-end Lean/Rocq certificate or an exported object-language ZFC derivation: it inherits the named unbounded calibration interfaces, their ordinary proofs, and their existing formal verification limitations. A finite replay cannot replace those interfaces.

2.6 Exact degree and operation count

Give each quantified coordinate degree one and the fixed ports degree zero. The six residual degrees in Equation (6) are

$$(34, 14, 5, 14, 10, 44).$$

In particular, the rank row rises from degree four to degree five after the projection; this does not change the dominant degree. Write X_*, Y_*, E_*, η_* for the top homogeneous parts. They are

$$X_* = 36\beta J_3 w, \qquad Y_* = 54\beta^3 J_3^3 v,$$
$$E_* = X_* Y_*, \qquad \eta_* = -hX_* Y_*^2.$$

Thus T has degree 22 and top part $-\eta_*^2$, whereas d has degree at most four. Only L_6^2 reaches degree 88, with top part η_*^8. In particular

$$[h^8 w^8 v^{16} J_3^{56}] Q_{18} = 2^{32} 3^{64} \beta^{56} > 0. \tag{9}$$

This proves exact degree on the actual fixed-coefficient slice, rather than merely giving a degree propagated through a circuit.

The delivered DAG uses 54 multiplications, 27 additions, and 24 subtractions. Every square and every scalar multiplication is charged; fixed numeral leaves and reuse of a register are free. The six residuals need 94 operations, and six squarings plus five additions give 105. The predecessor needs 109. Restoring F costs additional arithmetic, but eliminating its transport row and shortening the rank expression save four live operations overall. No minimum-operation claim is made.

2.7 Three companion tradeoffs

The same transformation applies to the incoming 20/32, 21/20, and 22/16 joint-sign polynomials. Their three resulting packets have, respectively, nineteen, twenty, and twenty-one unrestricted coordinates. They retain some intermediate coordinates that Equation (7) eliminates. Precisely:

- The nineteen-variable, degree-32 form retains η and the row $c - \eta - kY = 0$.
- The twenty-variable, degree-20 form also retains E and the row $E - XY = 0$, while $a = E + Y$ remains an abbreviation.
- The twenty-one-variable, degree-16 form also retains a and $a - E - Y = 0$. Its source rows use $a - E$ in place of Y where specified by the parent. It retains the parent's reduced auxiliary residual $\Delta(f^2 - 1)(V^2 - y^2) + y^2 - 1$.

The last replacement is justified in the parent by retaining $I^2 - \Delta(f^2 - 1)$ and subtracting its multiple $V^2 - y^2$ from the original auxiliary row. It is a zero-set-preserving row operation, not an unrestricted polynomial identity. Our new substitution preserves the particular parent polynomial for each endpoint.

Their independently checked degree certificates are

$$[\eta^4 h^4 w^4 v^4 J_3^{16}]Q_{19} = 2^{12}3^{20}\beta^{16},$$
$$[f^4 v^4 J_3^{12}]Q_{20} = 2^4 3^{12}\beta^{12},$$
$$[a^4 f^8 o^4]Q_{21} = 1.$$

All are positive for the authentic tuple. The expanded row degrees and literal circuit counts are in the verification receipts. These exact integer-domain improvements coexist with lower-variable, much-higher-degree equations; no scalar measure makes every tradeoff comparable.

2.8 Reproduction and provenance

The component commands are `python3 build.py`, `python3 verify.py`, and `python3 -O verify.py`. The independent checker does not import the new builder. It authenticates the retained source files and coefficient tuple, rechecks the parent DAGs against their literal polynomials, proves every new residual identity by exact sparse algebra over validated cut registers, checks degree bounds and nonzero top coefficients, verifies the sum-of-squares finalizer, and rejects an operation-preserving sign corruption. Normal and optimized Python runs pass. No diagnostic search for a consistency-equation zero is used.

The primary polynomial is `transport18_degree88.json`; the three companions are `transport19_degree32.json`, `transport20_degree20.json`, and `transport21_degree16.json`. Each declares every witness, fixed numeral, gate, residual, and restoration map. The inherited proof chain, source 73's earlier method, literal finite recipes, fixed bytes, licenses, and source hashes are retained under `upstream/`. The method is credited to that earlier report; the current contribution is its exact composition with the newer joint-sign family.

3 Removing a redundant Pell factor

The central improvement replaces a square of an already certified square by that square itself. The square root needed to explain the replacement exists only in the proof; it does not become an additional unknown. This changes the arithmetic certificate, so it requires a new soundness argument. It produces two complete division-free polynomial circuits:

Construction	Positive unknowns	Exact degree	Operations
Frozen H2082 shared-product parent	9	1,845,940	440
Saturated certificate, simple bound	9	1,628,768	434
Saturated certificate, whole radical	9	1,103,950	1,074

Operations are live binary additions, subtractions and multiplications; squares and scalar products are charged, and fixed numeral leaves and register reuse are free. The natural-number translations use respectively 441 and 1,081 operations, with unchanged degrees. These are two different tradeoffs, rather than a claim that either is globally simplest.

3.1 The new Pell representation

Let $\psi_A(0) = 0$, $\psi_A(1) = 1$ and $\psi_A(t+2) = 2A\psi_A(t+1) - \psi_A(t)$. Let χ_A be the companion sequence with initial values $1, A$. Thus

$$\chi_A(t)^2 - (A^2-1)\psi_A(t)^2 = 1.$$

For $A \geq 2$, the positive Pell solutions are exactly these pairs.

Theorem 3.1 (Saturated positive Pell certificate)**.** *Let A, B, C, J be positive integers satisfying $A \geq 3$, $1 < B < A$, $C > B$ and $J \geq A$. For $i \geq 1$ and $j_0 \geq 0$ put*

$$\begin{aligned} D &= 1 + (A^2-1)C^2, & e_0 &= iJC^2, \\ F &= 1 + (A^2-1)De_0^2, & \Lambda &= CD + 1 - A, \\ G &= A + F\Lambda, & H &= C + F(B + j_0C), \\ I &= 1 + (G^2-1)H^2. \end{aligned}$$

Then $C = \psi_A(B)$ if and only if some such i, j_0 make DFI a square.

Proof. All displayed radicands and G, H are positive. In particular $D \geq A^2$ and $\Lambda \geq A^2 - A + 1 > 0$. The congruences

$$\begin{aligned} F &\equiv 1 \pmod{D}, & G &\equiv 1 \pmod{D}, & I &\equiv 1 \pmod{D}, \\ G &\equiv A \pmod{F}, & H &\equiv C \pmod{F}, & I &\equiv D \pmod{F} \end{aligned}$$

show that D, F, I are pairwise coprime. If their product is a square, each factor is a square. Set $d = \sqrt{D} \in \mathbb{Z}_{>0}$. Then

$$F = 1 + (A^2-1)(iJC^2d)^2.$$

Pell classification gives positive indices s, m, t such that

$$C = \psi_A(s), \qquad iJC^2d = \psi_A(m), \qquad H = \psi_G(t).$$

The standard rank argument gives $C \mid m$. Here are its essential details, including the case of even C. From $\psi_A(s) \mid \psi_A(m)$ the divisibility law gives $s \mid m$; write $m = su$. Expanding the Pell addition identities yields

$$\psi_A(su) \equiv u\chi_A(s)^{u-1}\psi_A(s) \pmod{\psi_A(s)^3}.$$

Since $\gcd(\chi_A(s), \psi_A(s)) = 1$, the stronger divisibility $\psi_A(s)^2 \mid \psi_A(m)$ implies $\psi_A(s) \mid u$, hence $C \mid m$. The usual divisibility law itself follows by Euclidean reduction of the addition identities and strict growth of ψ_A.

Because $G \equiv A \pmod F$ and $H \equiv C \pmod F$, recurrence reduction gives

$$\psi_A(s) \equiv \psi_A(t) \pmod{\chi_A(m)}.$$

This implies $s \equiv \pm t \pmod{2m}$. A short proof avoids any exceptional-index assumption. Put $x = \chi_A(m)$. The addition identities give

$$\psi_A(v + 2m) \equiv -\psi_A(v), \qquad \psi_A(2m - v) \equiv \psi_A(v) \pmod x.$$

Every index folds to $r \in [0, m]$, with residue $\pm\psi_A(r)$ and index congruent to $\pm r \pmod{2m}$. The Pell identity and $A \geq 3$ imply $x > 2\psi_A(m)$. The signed representatives therefore lie strictly between $-x/2$ and $x/2$. Equal residues are equal integers, and strict growth forces equal folded indices.

On the other hand $G \equiv 1 \pmod C$ and $H \equiv B \pmod C$, so $t \equiv B \pmod C$. Therefore $s \equiv \pm B \pmod C$. Since $C > 1$, we have $s \geq 2$; the inequality $\psi_A(s) \geq As$, together with $B < A$, gives $0 < s + B < C$. Consequently the only possibility is $s = B$.

Conversely, suppose $C = \psi_A(B)$ and put $d = \chi_A(B)$, so $d^2 = D$. The determinant-one matrix

$$M_A = \begin{pmatrix} 2A & -1 \\ 1 & 0 \end{pmatrix}$$

has finite order modulo JdC^2. Choose a positive multiple m of this order and set $i = \psi_A(m)/(JdC^2) > 0$. Then $F = \chi_A(m)^2$. Let T be a positive period of M_G modulo F. Choose arbitrarily large $t \equiv B \pmod{TC}$ and put $H_0 = \psi_G(t)$. We have

$$H_0 \equiv C \pmod F, \qquad H_0 \equiv B \pmod C,$$

using respectively $G \equiv A \pmod F$ and $G \equiv 1 \pmod C$. Since $F \equiv 1 \pmod C$, the quotient $(H_0 - C)/F$ is B modulo C. Taking t sufficiently large makes that quotient at least B; hence

$$j_0 = \frac{(H_0 - C)/F - B}{C} \in \mathbb{N}.$$

Then $H = H_0$ and $I = \chi_G(t)^2$. This supplies the required witnesses. All divisions here occur in the witness proof, not in the polynomial. □

The old certificate used

$$F_{\text{old}} = 1 + (A^2 - 1)D^2(i_{\text{old}}JC^2)^2.$$

An old valid witness gives a particularly simple new witness: $i = \sqrt{D}\, i_{\text{old}}$. It preserves F, G, H, I exactly. The converse is supplied by Theorem 3.1, rather than by an unjustified division of the free integer i by $\sqrt{D}$.

3.2 Binding to the actual compact54-sharp source

Both new equations retain the entire H2082 source prefix and its coherent compact54-sharp coefficient tuple from commit `c1a102936aca437e9999530a0111e6ddab56dc6d`. The complete layout, three kernel polynomials and coefficient leaves are included as literal data. The four original coefficient ports occupy

215,492 raw bytes; the eleven expanded live fixed leaves occupy 1,565,018 raw bytes. These are different representations of the fixed data and must not be confused with the number of unknowns.

For the inherited two-block words $S_{\mathrm{code}}, T_{\mathrm{code}}$, define

$$\begin{aligned} B_{\mathrm{code}} &= 2^{864026}b^2, & L &= bB_{\mathrm{code}}^{2200}, \\ N &= 2^{6480184}B_{\mathrm{code}}^{1162}, & R &= L(S_{\mathrm{code}} + T_{\mathrm{code}} + 1) - S_{\mathrm{code}} - 1, \\ t_{\mathrm{cap}} &= bB_{\mathrm{code}}^{116}. \end{aligned}$$

The supplied source tables define both words without a free program index. The inherited source theorem gives $R, N \geq 8$ and the required carry/binomial calibration. Define the ratio registers

$$\begin{aligned} U &= N^2w, & Y &= N^2s, & M &= RY, \\ P_0 &= 2M^2U, & K &= R + 1 + h(P_0 - 1), & A &= M(U + 1), \\ B_1 &= 2R + 1, & W_1 &= bw, & J &= 4A - 5, \\ Q_0 &= 3W_1C - 2(W_1^2 - 1), & & & & \\ a_0 &= 1 + (P_0^2 - 1)K^2, & V_0 &= P_0K, & & \\ m_0 &= K^2(4t_{\mathrm{cap}} - g) - 16t_{\mathrm{cap}}(C - KY)^2, & & & & \\ \Omega &= Jm_0, & E_0 &= J(n + W_1^2) + Q_0. & & \end{aligned}$$

The nine inputs are

$$b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}.$$

Set $j_0 = j - 1$ and use $B = B_1$ in Theorem 3.1. The source bounds imply $A \geq 33280$, $1 < B_1 < A$ and $J \geq A$ on every ambient positive assignment. The hypothesis $C > B_1$ will be established from the final equation before the representation theorem is invoked.

3.3 The 434-operation equation

Define

$$T_0 = ADe_0, \qquad V_1 = T_0GH, \qquad a_1 = DFI, \qquad \Theta = E_0 - \Omega(V_0 + V_1).$$

The first new equation is

$$\boxed{\left(\Theta^2 - \Omega^2(a_0 + a_1)\right)^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{10}$$

Its unconditional majorant is exceptionally small:

$$\begin{aligned} T_0^2 - DF &= D(De_0^2 - 1) > 0, \\ V_1^2 - a_1 &= (T_0^2 - DF)(GH)^2 + DF(H^2 - 1) > 0. \end{aligned} \tag{11}$$

Thus $V_1 > \sqrt{a_1} > 0$. Also $V_0^2 - a_0 = K^2 - 1 > 0$. These inequalities hold before a solution has been assumed.

Proposition 3.2. *Equation (10) has a positive solution exactly when the same pinned native source has a solution.*

Proof. The identity

$$E_0 = Jn + (J - 2)W_1^2 + 3W_1C + 2 > 0$$

shows that a zero cannot have $m_0 = 0$, since then its left side would be E_0^4. Factoring the two-root norm over the reals gives signed roots $r_0^2 = a_0$, $r_1^2 = a_1$ satisfying

$$\frac{E_0}{J} = m_0(V_0 + V_1 + r_0 + r_1). \tag{12}$$

The strict majorants make the parenthesis positive for every sign choice; therefore $m_0 > 0$ before any square-root integrality is used.

The unchanged margin argument now gives $|C - KY| < K/2$ and $C > B_1 + W_1$. Moreover $A^2 > P_0$, $D > C^2$, $F \geq A^2$ and $I \geq A^2$, so

$$DFI > C^2A^4 > K^2P_0^2 > a_0.$$

Hence the rational sum $x = \Theta/\Omega = r_0 + r_1$ is nonzero. The identity $2xr_0 = x^2 + a_0 - a_1$ makes each root rational, hence integral. Equation (12) then gives $J \mid Q_0$. Theorem 3.1 gives $C = \psi_A(B_1)$. The retained arithmetic ratio/binomial theorem recovers the source zero. If its original auxiliary certificate is desired explicitly, the finite period argument in Theorem 3.1 can be repeated using the modulus JDC^2 instead of $J\sqrt{D}C^2$.

Conversely, use the inherited canonical ratio witnesses from a source zero. Set $i = \sqrt{D}\, i_{\mathrm{old}}$ to preserve the old F, G, H, I, keeping all other ratio coordinates fixed. Choose positive integral roots and put

$$n = m_0(V_0 + V_1 + \sqrt{a_0} + \sqrt{a_1}) - W_1^2 - Q_0/J.$$

This is an integer. The retained bound $W_1^2 + Q_0/J < 2D$, together with $V_1 > 2D$ from $T_0 = ADe_0 > 2D$, proves $n > 0$. One norm factor vanishes. □

Composing this existential equivalence with the named source/refutation calibration, and then negating, gives the intended ordinary argument in ZFC. The new proof does not by itself close the repository's remaining internal-model calibration or produce a new kernel-checked object-language consistency biconditional.

3.4 A smaller-degree majorant of the whole radical

Temporarily treat A, C, i, Z as independent indeterminates, put $J = 4A - 5$, and replace H by $C + FZ$. Set

$$\mathcal{Q} = DF\big((G^2 - 1)H^2 + 1\big).$$

Its A-degree is 36, with leading coefficient $(1024C^{19}i^5Z)^2$. Write $\mathcal{Q} = \sum q_k A^k$ and define

$$t_{18} = 1024C^{19}i^5Z,$$
$$t_{18-k} = \frac{q_{36-k} - \sum_{\ell=1}^{k-1} t_{18-\ell}t_{18-k+\ell}}{2t_{18}} \qquad (1 \leq k \leq 18).$$

This finite calculation takes place in the coefficient field $\mathbb{Q}(C, i, Z)$. It specifies the polynomial part of the formal square root; it is not a division performed on numerical witnesses. Exact arithmetic shows that

$$\sigma = 1024C^5 i, \qquad \mathcal{P} = \sigma \sum_{r=0}^{18} t_r A^r$$

has 333 integer monomials and no negative exponents. Define

$$\mathcal{L} = 2^{20}C^{24}i^6Z, \qquad \mathcal{P}_+ = \mathcal{P} + \mathcal{L}, \qquad \mathcal{D} = \mathcal{P}_+^2 - \sigma^2\mathcal{Q}. \tag{13}$$

The complete coefficient lists are supplied. Two exact integer-polynomial identities give the uniform inequalities:

- $\mathcal{P}(A+3, C+1, i+1, Z+1)$ has 6,650 strictly positive coefficients, with constant 15 915 642 730 652.
- The translated defect $\mathcal{D}(A+3, C+1, i+1, Z+1)$ has 36,309 strictly positive coefficients. Its constant is 32 237 226 735 615 909 648.

Consequently

$$\mathcal{P}_+(A, C, i, Z) > \sigma\sqrt{\mathcal{Q}} > 0 \quad (A \geq 3, C, i, Z \geq 1). \tag{14}$$

This is a universal positivity certificate. The uncorrected formal polynomial part would be invalid: its defect at $(3, 1, 1, 1)$ is $-1\,140\,296\,347\,768\,021\,232$. The correction in (13) is therefore substantive.

For the second fixed equation put

$$Z = B_1 + (j-1)C, \quad V_1 = \mathcal{P}_+(A, C, i, Z), \quad a_1 = (1024C^5 i)^2 DFI,$$

and use exactly the norm equation (10). The previous proof applies with one additional observation: once the signed root r_1 is integral, r_1/σ is a rational square root of the integer DFI, hence is itself an integer. Thus the scale introduces no extra square solutions. In the converse direction $F > D$ and $\sigma \geq 1024$ imply $V_1 > 2D$, so the witness n remains positive.

The supplied graph evaluates the 333 coefficients by fixed polynomial operations. All 24 multivariate Horner orders were counted; the selected order is Z, i, C, A. This yields 1,074 live binary operations. It is a finite evaluated circuit count, not a claim of an optimal arithmetic circuit.

3.5 Exact degrees and the changed dominant term

The source has

$$r = \deg R = 8799, \qquad \nu = \deg N = 2324, \qquad W = 116, \qquad a = r + 4\nu + 2 = 18097.$$

The common register degrees are

$$\begin{aligned} \deg \Omega &= 5a + 2W + 3 = 90720, \\ \deg V_0 &= 3a + r + 1 = 63091, \\ \deg \delta_0 &= 3a + r + 2 = 63092, \qquad \delta_0 = K^2 - 1, \\ \deg E_0 &= a + 4 = 18101, \qquad \deg a_0 = 126182. \end{aligned}$$

For either majorant write $\delta_1 = V_1^2 - a_1$ and

$$B_* = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1).$$

The universal exact identity

$$F_9 = B_*^2 + 4\Omega^2 V_0 V_1 B_* + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1) \tag{15}$$

resolves the leading cancellations that a raw syntactic degree count misses.

For the simple majorant,

$$\deg V_1 = 18a + r + 24 = 334569, \qquad \deg \delta_1 = 34a + 2r + 48 = 632944.$$

The latter follows directly from (11). The square B_*^2 is uniquely dominant in (15); therefore

$$\deg F_9 = 92r + 352\nu + 8W + 284 = 1\,628\,768.$$

Its leading coefficient is positive on the actual source tuple.

For the complete-radical bound the unique greatest weighted monomials, with weights $(a, 1, 1, r)$ on (A, C, i, Z), are

$$2^{20}A^{18}C^{24}i^{6}Z \quad \text{in } \mathcal{P}_+, \qquad 2^{41}A^{18}C^{48}i^{12}Z^{2} \quad \text{in } \mathcal{D}.$$

Hence $\deg V_1 = 334575$, $\deg \delta_1 = 343404$ and $\deg B_* = 524844$. The four summands in (15), splitting its last parenthesis, have degrees

B_*^2	1,049,688
$4\Omega^2 V_0 V_1 B_*$	1,103,950
$4\Omega^4 \delta_0 V_1^2$	1,095,122
$4\Omega^4 a_0 \delta_1$	832,466

The second term is now uniquely dominant, with positive leading coefficient. Thus the exact degree is

$$\boxed{\deg F_9 = 63r + 236\nu + 8W + 221 = 1\,103\,950.}$$

Translation of positive inputs to natural successors preserves these degrees; the natural j directly supplies the existing $j - 1$ coordinate.

3.6 What the executable evidence proves

The independent verifier uses a separately implemented sparse integer polynomial ring and does not import either circuit builder or the rational discovery code. It checks the entire radical identity, shifted positive coefficient lists, all local certificate formulas, the exact final norm, source-prefix structural equality, topology, liveness, operation counts, all fixed-leaf hashes and the resolved degree comparisons. It rejects both the uncorrected radical truncation and a corrupted final norm. The complete coefficient generators and parent proofs are retained as explicit dependencies; no finite check is presented as a consistency test.

The main mathematical change is the saturation theorem, whose soundness and finite-period completeness were proved above. The whole-radical certificate is a second new uniform lemma. The remaining native/compiler and syntactic-refutation interfaces are inherited, and their unresolved formal verification boundary remains visible.

4 A different source: fixed coprime formula labels

This section supplies a new source-level alternative to the affine-factor and coefficient-budget sources inspected at commit `c1a102936aca437e9999530a0111e6ddab56dc6d`. It does not supply a smaller completed Diophantine equation. The change removes a freely chosen common multiple or a coefficient-height budget from the source; the cost of computing and retrieving its labels remains to be compiled. The closure argument itself is the single-conclusion Horn-flow lemma already used by package 68. The different ingredient here is a fixed, parameter-free sequence of pairwise coprime labels.

4.1 Coprime labels do not need to be primes

For $j \in \mathbb{N}$ put

$$F_j = 2^{2^j} + 1.$$

The elementary identity

$$\prod_{i<j} F_i = F_j - 2$$

follows by induction, including the empty product at $j = 0$. Consequently $F_j > 1$ and the F_j are pairwise coprime: a common divisor of F_i, F_j, with $i < j$, divides F_j and $F_j - 2$, and is odd. Primality is unnecessary; $F_5 = 641 \cdot 6700417$ is allowed.

For any finitely supported vectors $a, b \in \mathbb{N}^{(\mathbb{N})}$,

$$\prod_j F_j^{a_j} = \prod_j F_j^{b_j} \quad \Longleftrightarrow \quad a = b. \tag{16}$$

To prove the nontrivial implication, cancel the minimum of the two exponents at every index. If a positive exponent remains on one side, its factor $F_j > 1$ is coprime to the whole other side, a contradiction. This proof requires no factorization algorithm or primality predicate.

4.2 An exact unordered proof certificate

Let $\mathcal{R}$ be any family of finitary Horn rules with natural-number labels, one conclusion c_r and a finite premise multiset P_r. Zero-premise rules are axioms. Let D be the least rule-closed set. Then for each target t the following conditions are equivalent:

1. $t \in D$.
2. Some finite list of valid rule instances satisfies

$$\prod_{i<N} F_{c_i} = F_t \prod_{i<N} \prod_{p \in P_i} F_p. \tag{17}$$

3. Some finite list of valid rule instances and natural numbers $S_0, \ldots, S_N$ satisfy

$$S_0 = F_t, \qquad S_N = 1, \qquad S_{i+1} F_{c_i} = S_i \prod_{p \in P_i} F_p. \tag{18}$$

For completeness, expand a finite proof tree root first. Let S_i be the product of the weights of the current open obligations. Expanding a node replaces its conclusion by its actual premises, giving (18); the empty frontier has product one.

For soundness of the product certificate, (16) gives the coordinate balance

$$\sum_{i<N} \left(e_{c_i} - \sum_{p \in P_i} e_p \right) = e_t.$$

If $t \notin D$, sum the coordinates outside D. Each rule contributes at most zero: if its conclusion is outside D, at least one premise is outside D, while a conclusion inside D contributes zero minus a nonnegative number. The right side is one, a contradiction. Thus cycles cannot create a false proof. A disconnected cycle can be present in a valid certificate, but it is harmless. Saturating the finitely many selected rules gives a finite proof DAG for the target; this can be unfolded to a proof tree.

The accumulator conditions imply the product certificate by multiplying and cancelling. All S_i are positive: this follows inductively from $S_0 > 0$ and positivity of every factor in (18). No independent positivity guard is needed when the accumulators range over naturals. We do not claim that an arbitrary ordering of an arbitrary balanced multiset gives an integral accumulator path. Existence is enough: the recovered proof tree provides another valid list and path if necessary.

Exactly one conclusion and nonnegative rule multiplicities are essential. The rule $a \longrightarrow \{a, t\}$ has balance e_t without any axiom. Also, subtracting the vector for $a, a \longrightarrow b$ from that for $a \longrightarrow b$ gives e_a, although neither rule derives anything from the empty set. Those are excluded by the stated hypotheses.

For the repository's target-one calculus, use the actual arity of each rule, so that truth code zero is a genuine zero-premise axiom. Then $F_0 = 3$ and the initial target weight is the fixed integer $F_1 = 5$. One must not simultaneously use $F_0 = 3$ and treat padded truth premises as free factors. A different, also sound convention sets the zero-label weight to one and retains Fermat weights only for positive labels, but requires a separate zero-label branch. The convention here avoids it.

4.3 A precise four-parameter bounded source

The double exponent in F_j is not an ordinary polynomial. To make its remaining compilation cost concrete, define

$$\beta(x, y, j) = x \bmod (1 + (j + 1)y).$$

Use four outer natural parameters N, x_S, x_Q, y, and put $S_i = \beta(x_S, y, i)$, $Q_i = \beta(x_Q, y, i)$. Impose the endpoints

$$S_0 = 5, \qquad S_N = 1, \qquad Q_0 = 2.$$

For every $i < N$, require simultaneously

$$Q_{i+1} = Q_i^2$$

and one actual rule instance $P_i \longrightarrow c_i$, all of whose labels are strictly less than N, with

$$S_{i+1}(\beta(x_Q, y, c_i) + 1) = S_i \prod_{p \in P_i} (\beta(x_Q, y, p) + 1).$$

Allow identity rules $c \longrightarrow c$ as padding; their addition does not change derivability. These conditions use one bounded universal block, with existential local witnesses. Induction gives $Q_j = 2^{2^j}$ for $0 \le j \le N$, so every referenced label is authenticated. The preceding theorem proves soundness.

For completeness, choose a finite proof tree and then a horizon N larger than all labels and at least its number of nodes. Pad the rule list by identities, keeping the terminal accumulator equal to one. Form the finite S and Q sequences first. Choose

$$y = N! \left(1 + \max_{0 \le i \le N} \{S_i, Q_i\}\right).$$

The moduli $1 + (i + 1)y$ are pairwise coprime and exceed the desired entries. The finite Chinese remainder theorem supplies x_S, x_Q with the same scale y. This factorial is used only to prove existence of CRT witnesses; it is not a function in the proposed final polynomial, and no source condition asserts that an unknown is a factorial. The choice order avoids any self-referential size bound.

4.4 What must still be charged

The preceding four-parameter source is a finite first-order arithmetic formula, not an ordinary fixed-dimensional polynomial. Its costs include the supplied Horn-rule recognizer, the two adjacent sequence entries, up to three indexed label lookups per binary rule, label-range witnesses, and the bounded-universal compiler. For example a lookup $u = \beta(x_Q, y, p)$ has the completely explicit quadratic graph

$$d = (p + 1)y, \qquad x_Q = (d + 1)k + u, \qquad u + s = d$$

over naturals. These are three equations using four local witnesses d, k, u, s, one of which is the returned value u. A strict label bound is $p + b + 1 = N$ with one further natural witness b. Shared denominators and fixed zero-premise branches may save registers, but no unmeasured savings are assumed.

Alternatively, one can define a label by two separately compiled exponentiation graphs. That removes the power sequence and introduces the corresponding graph witnesses. Both routes remove the source's common multiple or coefficient budget; neither has yet been shown to reduce the complete polynomial's variable count, degree or arithmetic cost. The finite script `alternative/verify.py` checks all rule-multiset cases listed in `alternative/VERIFICATION.json`, constructs proof trees and integer traces, verifies a small shared-CRT example, and includes the failure modes above. It is an adversarial regression, not a substitute for these universal proofs or the inherited calculus-to-ZFC calibration.

5 Reproducibility, source binding, and integration

5.1 What the archive contains

The main deliverables are this article and its editable LaTeX source. The `integer/` directory contains four explicit equations, paid circuits, six fixed coefficient payloads and their recipes, source copies, the inverse maps, and independent polynomial checks. The `nine/` directory contains both positive constructions, natural translations, the complete fixed-leaf binding, source/layout/kernel data, and the radical-majorant certificates. The `alternative/` directory contains the separate finite proof-flow construction and its diagnostic examples. `provenance/BASELINE.json` pins the reviewed commit and all five incoming ZIPs by SHA-256 and Git blob identifier. Component notices retain the inherited licenses and attribution.

Use Python 3.10 or newer for the standard-library checks:

```
python3 verify.py
```

The verification receipt distinguishes exact algebra from finite diagnostics and the inherited unbounded proof dependencies. The optional discovery calculations are separate from checking the released polynomial and its certificates. The full coefficient bytes are included, so validation does not require a network connection or a live GitHub branch.

To rebuild this PDF with a standard LaTeX installation:

```
make pdf
```

To rebuild the release ZIP and its content manifest:

```
python3 make_archive.py
```

5.2 What should be incorporated into ProveIt

The most direct new candidate is the 18-integer degree-88 equation: its proof interface is a short polynomial automorphism and monic elimination. It improves a recently delivered low-degree branch without changing its coefficient tuple. The three lower-degree companions should be retained with it rather than choosing a single point without a cost criterion.

For the nine-positive branch, the saturated short-majorant circuit improves both cost and degree over the current 440-operation candidate. The complete-radical form should be retained as a second candidate when total degree is the priority. Its large positivity certificate is a finite proof object that can be checked independently; it should not be confused with a short hand-expanded formula or a proof of the native compiler.

The direct proof-flow construction belongs with source-level alternatives. Its removal of a global proof-order certificate is established, but the lookup and aggregation costs must be compiled and counted before it can replace a finished consistency equation. These are concrete next obligations, not missing parameters hidden inside the reported metrics.

5.3 Limits of the result

No global simplicity minimum is proved. The fewest possible number of variables for a polynomial specific to this consistency statement is not determined. We do not infer historical novelty from the absence of a formula in the inspected snapshot. Large fixed numerals, high total degree, short arithmetic circuits, and low witness counts remain different objectives.

No zero search or consistency assertion is part of verification. The equations are designed so that any zero provides a finite refutation and any finite refutation supplies a zero through the stated proof chain. Their mathematical significance is the biconditional, including the reverse construction, not an empirical failure to find a solution.

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